

Cross-validation

Dataset splitting functions.

```
lightfm.cross_validation.random_train_test_split(interactions,  
test_percentage=0.2, random_state=None) \[source\]
```

Randomly split interactions between training and testing.

This function takes an interaction set and splits it into two disjoint sets, a training set and a test set. Note that no effort is made to make sure that all items and users with interactions in the test set also have interactions in the training set; this may lead to a partial cold-start problem in the test set. To split a `sample_weight` matrix along the same lines, pass it into this function with the same `random_state` seed as was used for splitting the interactions.

Parameters:

- **interactions** (*a scipy sparse matrix containing interactions*) – The interactions to split.
- **test_percentage** (*float, optional*) – The fraction of interactions to place in the test set.
- **random_state** (*int or numpy.random.RandomState, optional*) – Random seed used to initialize the `numpy.random.RandomState` number generator. Accepts an instance of `numpy.random.RandomState` for backwards compatibility.

Returns: (**train, test**) – `scipy.sparse.COOMatrix` A tuple of (train data, test data)

Return type: (`scipy.sparse.COOMatrix`,